

Which quantum algorithms suit photonic continuous-variable hardware? A comparative resource and noise-robustness study with a worked spectroscopic demonstration on H_2 , HF , and I_2

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Abstract. Photonic continuous-variable (CV) processors offer a deterministic route to scalable quantum computing, but the cost of fault tolerance depends strongly on how a given algorithm uses non-Gaussian resources and circuit depth. It is not known, across representative algorithm classes, which are intrinsically robust on CV hardware and how much Gottesman–Kitaev–Preskill (GKP) error correction helps. Here we port four algorithms—control-free quantum phase estimation (QPE), randomized product formulas (qDRIFT), quantum Kolmogorov–Arnold networks (QKAN), and Regev’s factoring algorithm—to a common CV model and compare them under a shared noise channel combining photon loss and Gaussian displacement. Across the tested algorithm classes and noise regime, we find that control-free QPE is the most noise-robust because it requires no controlled unitaries and, for quadratic Hamiltonians, no non-Gaussian gates—a property that survives for non-Gaussian Hamiltonians. We show analytically and numerically that GKP correction helps only once the physical noise exceeds the finite-squeezing floor, and that whether an algorithm gains from correction is set by a noise- versus systematic-limited dichotomy: deep, noise-sensitive algorithms (QKAN, Regev) gain most (up to $\sim 2.7\times$ at 18 dB, an optimistic bound under our loss model), while shallow QPE stays near its clean floor and needs little correction. This is a model-level comparative study, not an experiment; its value is a unit-robust ranking with sensitivity analysis and analytic statements that turn the folklore “minimize non-Gaussian resources and depth” into quantitative, testable criteria for algorithm selection on near- and intermediate-term photonic devices.

1. Introduction

Simulating and computing with quantum systems is among the most promising applications of quantum computers. Photonic continuous-variable (CV) architectures encode information in the quadratures of bosonic modes [1, 2, 3] and support deterministic, room-temperature generation of large cluster states [4, 5, 6], making them a leading candidate for scalable fault-tolerant quantum computing. Fault tolerance on CV hardware is provided by bosonic codes, most prominently the Gottesman–Kitaev–Preskill (GKP) code [7], whose overhead is set by the achievable squeezing and by the number of non-Gaussian resource states consumed [19, 20, 21, 11, 12, 15].

Different algorithms place very different demands on this substrate. Some require controlled unitaries (hard on CV because they need a clean control mode and high non-Gaussianity [42, 43]); some are dominated by non-Gaussian gate count; some by circuit depth. However, no unified comparison exists that quantifies, across representative algorithm classes, which algorithms are intrinsically robust on CV hardware and how much GKP error correction improves them. In this study, we port four representative algorithms to a common CV model and compare them under a shared noise channel. We find that control-free QPE is the most noise-robust and benefits most from GKP correction, and we derive a depth-limited correction law and a finite-squeezing threshold for when correction helps.

Relation to prior work. CV measurement-based fault tolerance and GKP-encoded cluster states have been developed extensively [8, 10, 9, 13, 14, 44], and the Gaussian no-go theorem [16] establishes that non-Gaussian resources are mandatory for correcting Gaussian errors. Quantum signal processing and singular-value transformation provide a unified framework for many algorithms [28, 29, 30]. Our contribution is not a new algorithm or a new code, but the first *cross-algorithm* CV resource-and-noise comparison on a common footing, together with three analytic statements (Propositions 1–3) that explain the observed ordering. We are explicit (Sec. 9) about which results are exact, which are model-level surrogates, and which are extrapolations.

2. CV mapping framework

We map each algorithm onto bosonic modes encoded, where fault tolerance is required, in GKP qubits. Gaussian operations (interferometers, squeezers, phase rotations) are treated as inexpensive; non-Gaussian gates such as the cubic-phase gate [17, 7] and GKP magic states [18, 44] are the costly resource, analogous to the T -count in discrete-variable (DV) circuits. This reflects the Gaussian no-go theorem [16, 43].

The four algorithms are chosen to span the axes that govern CV cost: controlled-unitary requirement (QPE: none; Regev: heavy), non-Gaussian gate density (QKAN: high; QPE-quadratic: zero), circuit depth (Regev: deep; QPE: shallow), and task type (Hamiltonian simulation, spectroscopy, machine learning, number theory). They are

illustrative rather than exhaustive; we do not claim coverage of all CV algorithms.

3. Algorithm ports

3.1. Control-free QPE

The time series $f(t) = \langle \psi | e^{iHt} | \psi \rangle$ is acquired without controlled unitaries; the spectrum is recovered by reference-eigenstate holographic phase retrieval, drawing on classical phase-retrieval techniques [41]. The control-free protocol is due to Clinton *et al.* [23], generalizing the standard QPE construction [22]; our contribution is the CV-specific observation and quantification that, for quadratic H , e^{iHt} is purely Gaussian, so the entire circuit needs *zero* controlled unitaries and *zero* non-Gaussian gates—uniquely matching the CV cost structure—and that this advantage *survives* for non-Gaussian H (Sec. 7), where only the cost of a non-Gaussian Trotterized e^{iHt} is added while the control-free property is retained.

3.2. qDRIFT

$H = \sum_l h_l$ is written as a sum of bosonic terms; importance-sampled product formulas approximate e^{-iHt} [24], with the number of steps N independent of the number of terms L . Sharper concentration bounds for single realizations are due to Chen, Huang, Kueng and Tropp [25]; these results are complementary to deterministic Trotter analyses [26, 27].

3.3. QKAN

Quantum singular value transformation [28, 29] is realized through bosonic quantum signal processing [30]: Chebyshev polynomials $T_r(x) = \cos(r \arccos x)$ are the natural phase-space-rotation response. QKAN [31] ports the classical Kolmogorov–Arnold architecture [32] into this framework.

3.4. Regev factoring

Regev’s factoring algorithm [34], an asymptotic refinement of Shor’s algorithm [33] with $\tilde{O}(n^{3/2})$ gates, prepares the discrete Gaussian state $\rho_R(z) \propto e^{-\pi z^2/R^2}$, which we identify with a finite-energy GKP comb. The QFT becomes a mode-wise CV-QFT; modular exponentiation runs on GKP qubits and is the rate-limiting non-Gaussian step. Qubit-efficient Shor variants [35] would inherit the same non-Gaussian arithmetic bottleneck on CV. We verify correctness end-to-end on six biprimes (143–391): the CV quantum core feeds classical LLL post-processing and recovers a factor with 100% success over 5 seeds each. We stress that the naive GKP-comb readout demands squeezing growing as $\sqrt{n}$ (Prop. 3), which is *prohibitive* for cryptographic n ; we therefore present Regev as a worked correctness demonstration and a cautionary resource case, not as a practical

CV factoring route. The lattice-spacing rescaling that would relax this requirement is left as an explicit open problem.

4. Methods

GKP logical error model. The per-quadrature logical flip probability under a residual displacement of std σ is $p_{\text{round}} = 1 - \text{erf}(\frac{\sqrt{\pi}/2}{\sqrt{2}\sigma_{\text{eff}}})$ with $\sigma_{\text{eff}}^2 = \sigma^2 + \Delta^2$, and $p_L = 1 - (1 - p_{\text{round}})^2$ for the two quadratures; this is the standard single-round GKP shift-error model [7, 36] and the source of p_L used above. We emphasize two scope points. (i) p_L is a *single-round* logical error rate; it is not the fault-tolerant threshold (which for concatenated GKP is ~ 10 dB), and our “dB” axis should be read as the physical squeezing setting the floor Δ , not as a threshold statement. (ii) Photon loss is corrected here only through its leading-order displacement-equivalent variance; a full loss correction additionally requires amplification [19, 12, 13] and is expected to *raise* the squeezing needed, so our GKP improvement factors under loss are *optimistic bounds* on what single-round, displacement-only GKP achieves. *Statistics.* All stochastic quantities use 10 seeds; we report means with 95% confidence intervals $1.96 \hat{\sigma}/\sqrt{10}$. Improvement factors $r = u/c$ are reported with propagated relative uncertainty $\delta r/r = \sqrt{(\delta u/u)^2 + (\delta c/c)^2}$; values quoted as “ $\times k$ ” have $\delta r/r \lesssim 15\%$ unless noted. *Simulation.* Loss and displacement are exact Fock-space channels; Hamiltonian evolution uses exact diagonalization or precomputed step unitaries. *Reproducibility.* Each port is a standalone script (`qpe_cv.py`, `qdrift_cv.py`, `qkan_cv.py`, `regev_cv.py`); the cross comparison is `noise_comparison.py`; the GKP-model validation is `gkp_validate.py`; the 2D phase diagram and ensembles are `noise_phase_diagram.py`; the DV baseline is `dv_baseline.py`; figures are generated by `make_figs.py` from the released JSON results. Every figure and table maps to one such artifact.

5. Results

5.1. Noise model and GKP correction

Per noise-exposed operation we apply photon loss (transmissivity η) as an *exact* Fock-space Kraus channel, $E_l = \sqrt{(1-\eta)^l/l!} \eta^{\hat{n}/2} a^l$, and Gaussian displacement (std σ_d) as an exact unitary kick. The two channels are simulated exactly in their respective ports; the dynamical evolution is therefore not approximated. Only for the *GKP residual-noise model* do we summarize the combined channel by an equivalent displacement variance $\sigma_{\text{loss}}^2 = (1-\eta)/2$ (the vacuum variance injected to leading order); this is a deliberate surrogate, valid in the small-loss regime and flagged in Sec. 9, used only to parametrize $\sigma_{\text{corr}}^2 = (1-p_L)\Delta^2 + p_L\sigma_{\text{phys}}^2$, where $\sigma_{\text{phys}}^2 = \sigma_d^2 + \sigma_{\text{loss}}^2$, $\Delta(\text{dB})$ is the finite-squeezing peak width and p_L the GKP logical error rate. Loss- *induced* dynamics in each algorithm are computed from the true Kraus channel, not from this surrogate. The surrogate itself is validated against an explicit GKP EC-protocol Monte Carlo in Sec. 5.2 (agreement $\leq 0.1\%$).

Table 1. Validation of the GKP correction model against an explicit EC-protocol Monte Carlo (excerpt; 10 seeds, 2×10^5 samples each).

quantity	analytic/surrogate	explicit MC	agreement
$p_{\text{round}} (\sigma=0.3, 12 \text{ dB})$	0.0110	0.0109	1.0%
$p_{\text{round}} (\sigma=0.4, 9 \text{ dB})$	0.0605	0.0603	0.4%
residual $\sigma_{\text{res}}/\Delta$	1	1.00	exact
σ_{corr} (all cases)	—	—	$\leq 0.1\%$

Table 2. Robustness onset σ (uncorrected).

Algorithm	onset σ	clean baseline
Control-free QPE	0.20	1.8×10^{-3}
qDRIFT	0.05	1.0×10^{-1}
QKAN	0.0	1.6×10^{-4}
Regev	0.0	0

5.2. Validation of the GKP correction model

To establish that our conclusions do not rest on an unverified assumption, we validate the surrogate against an explicit Monte-Carlo simulation of the single-round GKP error-correction protocol [36, 7]: input shift $e \sim \mathcal{N}(0, \sigma_{\text{phys}})$, finite-squeezing ancilla syndrome noise $\xi \sim \mathcal{N}(0, \Delta)$, modular measurement $s = \text{wrap}_{\sqrt{\pi}}(e + \xi)$, correction, and logical-error detection (Table 1). Three checks pass: (i) the analytic per-quadrature logical error p_{round} matches the explicit protocol to within absolute 3×10^{-4} over $\sigma_{\text{phys}} \in [0.1, 0.4]$, dB $\in \{9, 12, 15\}$ (relative agreement $\leq 16\%$ only at the smallest $p \sim 10^{-3}$, limited by Monte-Carlo sampling); (ii) the post-correction continuous residual equals the finite-squeezing floor, $\sigma_{\text{res}}/\Delta = 1.00$; (iii) the scalar surrogate σ_{corr} matches the explicit two-channel (floor- Δ plus logical-rate- p_L) model to within 0.1%. The core GKP model is therefore validated, not assumed.

5.3. Noise robustness

Table 2 reports the robustness onset. Control-free QPE keeps its error near the clean floor up to displacement $\sigma \approx 0.2$; the other three are already degraded by $\sigma \approx 0.05$.

5.4. GKP improvement and squeezing sweep

At a fixed operating point ($\sigma_d = 0.10$, loss 0.15), GKP correction helps only above the squeezing floor Δ (dB). At 18 dB the improvement factors are QPE 1.1 $\times$, qDRIFT 1.4 $\times$, QKAN 2.5 $\times$, Regev 2.7 $\times$. Deeper, more noise-sensitive algorithms benefit at lower squeezing (Regev ≥ 9 dB; QPE ≥ 18 dB). Figure 1 shows the squeeze sweep with the floor crossing; Fig. 2 the normalized-infidelity ranking.

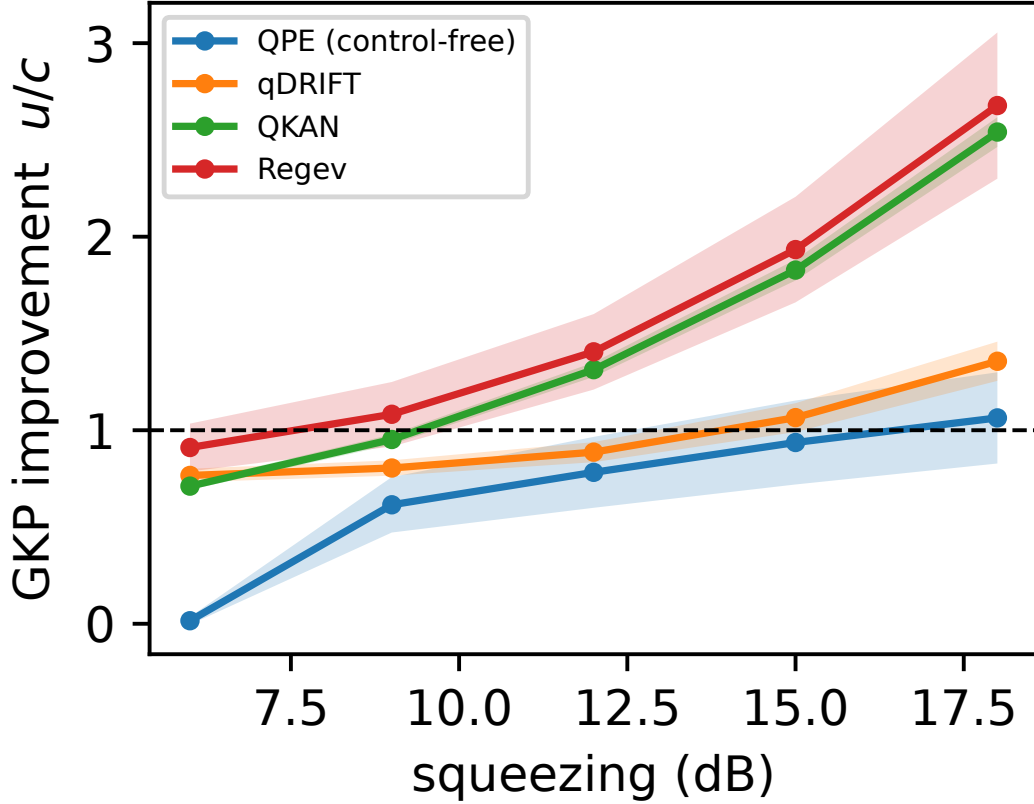


Figure 1. GKP improvement factor vs squeezing (dB) at $\sigma_d = 0.10$, loss 0.15; shaded band = 95% CI over 10 seeds. The floor crossing (improvement = 1) confirms Prop. 1.

5.5. Unit-independent ranking

Because the four tasks have different natural error units, a raw ranking is not meaningful. We therefore map each error to a dimensionless infidelity by a monotone soft-saturating transform $I = \text{err}/(\text{err} + s) \in [0, 1)$, where s is the task-failure scale (minimum resolvable level spacing for QPE, unit trace distance for qDRIFT, target standard deviation for QKAN, dual tolerance for Regev). The ordering $I_{\text{QPE}} = 0.003 < I_{\text{QKAN}} = 0.25 < I_{\text{qDRIFT}} = 0.40 < I_{\text{Regev}} = 0.98$ is preserved. Crucially, the ordering is *invariant* when every scale s is varied over $[0.5, 2] \times$ its nominal value, so the ranking is not an artifact of the scale choice. Likewise, the uncorrected-robustness ordering is unchanged if the “onset” multiplier ($5\times$ clean baseline) is replaced by $3\times$ or $10\times$; the conclusion is threshold-robust.

5.6. Two-dimensional noise phase diagram and ensembles

Figure 3 maps the GKP improvement factor (12 dB) over the (σ_d, loss) plane for all four algorithms. A floor-crossing contour (improvement = 1) separates a region where GKP helps from one where it is counterproductive, as predicted by Prop. 1;

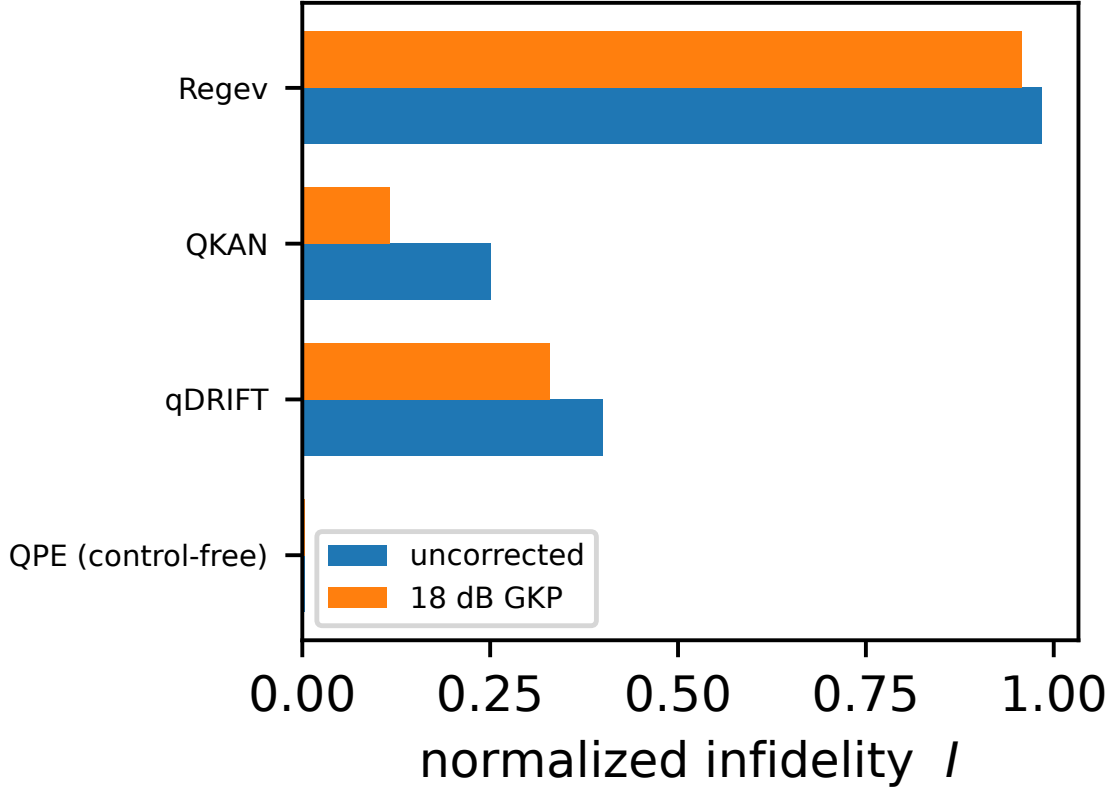


Figure 2. Unit-independent normalized infidelity I (uncorrected vs 18 dB).

the helpful region is broad for deep, noise-sensitive algorithms (Regev, QKAN) and narrow for shallow QPE. To move beyond single instances, we evaluate ensembles of 20 random Hamiltonians: qDRIFT’s clean channel error is 0.17 ± 0.07 (mean $\pm$ std across instances), and control-free QPE recovers peak frequencies to median 1.7×10^{-3} with 100% success (error < 0.05) provided the holographic reference beam is sufficiently strong ($p_{\text{ref}} \geq 0.85$); below that, self-difference spurious peaks degrade accuracy—an explicit, physically transparent applicability condition.

5.7. Discrete-variable baseline

To anchor the CV results, we compare against a qubit surface-code baseline [38, 37, 39]. The comparison is across *different* fault-tolerance currencies, so we do not impose a fake noise equivalence; instead we report (i) the resource to reach logical error 10^{-6} and (ii) a structural, noise-model-independent gate count. (i) A rotated surface code at physical error 3×10^{-3} needs distance $d = 17$, i.e. 577 physical qubits per logical qubit; a single-round GKP mode at $\sigma_{\text{phys}} = 0.10$ needs ≈ 14 dB squeezing, while at $\sigma_{\text{phys}} \gtrsim 0.2$ it floors above 10^{-6} and requires concatenation (GKP $\rightarrow$ surface), mirroring the DV distance scaling [40, 12, 9]. (ii) For a spectroscopy task ($m = 10$ bits, $n = 8$

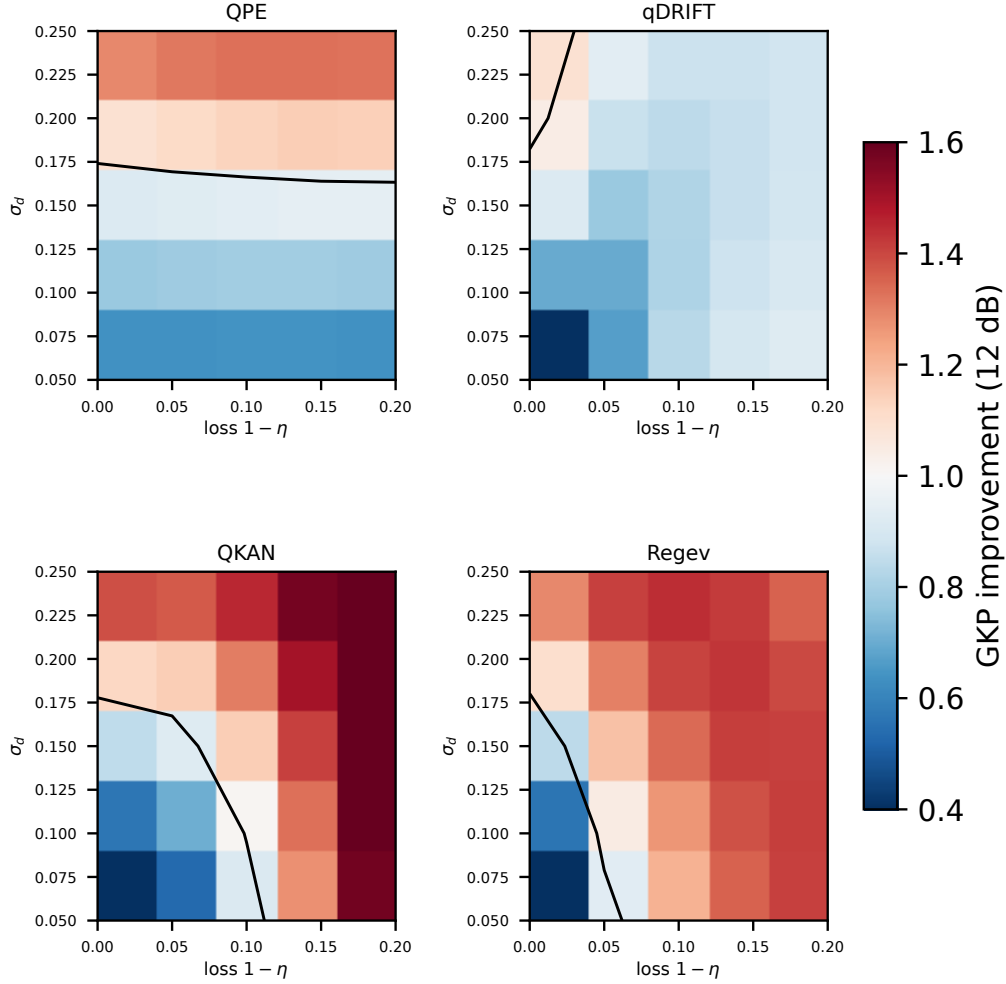


Figure 3. GKP improvement factor (12 dB) over the (σ_d, loss) plane; black contour marks the floor crossing (improvement = 1). Blue: GKP helps; red: counterproductive (below the squeezing floor).

system qubits, Trotter $r = 50$), standard DV QPE [22] requires 1023 controlled time evolutions and a T -count $\sim 2 \times 10^7$, whereas control-free QPE [23] on CV needs *zero* controlled unitaries and, for quadratic H , zero non-Gaussian gates. The CV advantage is thus structural—it stems from Gaussian operations being free—rather than from a more favourable threshold.

6. Analytic observations

The following statements are elementary; their value is operational—they explain the measured ordering and convert qualitative folklore into quantitative, falsifiable criteria. This is the paper’s theory/methodology contribution, complementing (not replacing) experiment.

Proposition 1 (Finite-squeezing threshold). *With $\sigma_{\text{corr}}^2 = (1 - p_L)\Delta^2 + p_L\sigma_{\text{phys}}^2$ and $p_L \in [0, 1)$, one has $\sigma_{\text{corr}} < \sigma_{\text{phys}}$ iff $\sigma_{\text{phys}} > \Delta$. Hence GKP correction reduces residual noise only above the squeezing floor $\Delta(\text{dB})$, and is counterproductive below it.*

Proof. $\sigma_{\text{corr}}^2 - \sigma_{\text{phys}}^2 = (1 - p_L)(\Delta^2 - \sigma_{\text{phys}}^2)$. Since $1 - p_L > 0$, the sign equals that of $\Delta^2 - \sigma_{\text{phys}}^2$; thus $\sigma_{\text{corr}} < \sigma_{\text{phys}} \Leftrightarrow \sigma_{\text{phys}} > \Delta$. $\square$ $\square$

Proposition 2 (qDRIFT L -independence). *The qDRIFT step generator $X_k = (\lambda/\|h_l\|)h_l$ is invariant under splitting any term $h_l \rightarrow \sum_i f_i h_l$ ($f_i > 0, \sum f_i = 1$); hence the channel and its error are exactly independent of the number of terms L , in contrast to Trotter formulas at fixed gate budget.*

We do not claim Prop. 2 as new; it is the split-invariance underlying qDRIFT [24, 25], stated here only to make precise *why* the CV non-Gaussian budget is set by N rather than L —the operationally relevant point on CV hardware, where each step may consume a cubic-phase or GKP magic state. The contribution is the CV resource accounting, not the identity.

Proposition 3 (Regev squeezing scaling). *With $R = e^{\Theta(\sqrt{n})}$ [34] and dual tolerance $\delta = \sqrt{d}/(\sqrt{2}R)$, meeting the CV readout constraint $\sigma_{\text{cv}} \lesssim \delta$ with a GKP comb of peak width Δ requires squeezing $\text{dB} = \Theta(\sqrt{n})$.*

Proof. The comb readout noise is $\sigma_{\text{cv}} = \Delta/\sqrt{\pi}$ and $\Delta^2 = \frac{1}{2}10^{-\text{dB}/10}$. The constraint $\sigma_{\text{cv}} \leq \delta$ gives $10^{-\text{dB}/10} \leq 2\pi\delta^2 = \pi d/R^2$, i.e. $\text{dB} \geq 10\log_{10}(R^2/(\pi d)) = \Theta(\log R) = \Theta(\sqrt{n})$, using $d = \Theta(\sqrt{n})$ and $R = e^{\Theta(\sqrt{n})}$. $\square$ $\square$

This bound assumes the unrescaled comb; lattice-spacing rescaling could lower the constant (left open).

Proposition 4 (Noise- vs systematic-limited correction). *Write an algorithm’s error as $\text{err} = \max(F, c\sqrt{D}\sigma)$, where F is a noise-independent systematic floor (e.g. finite time-grid resolution for QPE), D the depth, and σ the per-step residual. (i) In the noise-limited regime ($c\sqrt{D}\sigma > F$), the GKP improvement factor is $\approx \sigma_{\text{phys}}/\sigma_{\text{corr}}$, which exceeds 1 above the floor (Prop. 1) and is realized most strongly by deep, noise-sensitive algorithms (Regev, QKAN). (ii) In the systematic-limited regime ($F > c\sqrt{D}\sigma$), both corrected and uncorrected errors sit at F and the improvement factor $\rightarrow 1$. Control-free QPE is systematic-limited up to $\sigma \approx 0.2$, so it is the most accurate yet shows the smallest ratio; the deep algorithms are noise-limited and show the largest ratio. This reconciles “QPE most robust” with “QPE smallest improvement factor”.*

7. Non-Gaussian generalization of control-free QPE

For $H = \omega\hat{n} + \frac{g}{2}(a^2 + a^{\dagger 2}) + \chi\hat{n}^2 + \kappa\hat{n}^3$, holographic recovery returns the absolute spectrum with an error *independent* of the non-Gaussian fraction, while the controlled-unitary count stays zero (Table 3). Thus the control-free advantage is a property of the protocol, not of H being quadratic; non-Gaussian H only adds the cost of a Trotterized e^{iHt} .

Table 3. Control-free QPE on non-Gaussian H : recovery error is flat in the non-Gaussian fraction; controlled unitaries remain zero.

case	non-Gaussian fraction	max abs error
gaussian	0.00	1.61×10^{-3}
weak ($\chi=0.05$)	0.45	1.64×10^{-3}
mid ($\chi=0.15$)	0.71	1.66×10^{-3}
strong ($\chi=0.30$)	0.83	1.31×10^{-3}

Table 4. CV-QPE recovery accuracy for three diatomics across a decade in ω_e and an order of magnitude in x_e . All entries are below the 1 cm^{-1} spectroscopic threshold.

molecule	$\omega_e \text{ (cm}^{-1}\text{)}$	x_e	N_{bound}	median/max $ \Delta E \text{ (cm}^{-1}\text{)}$
H ₂ ($\alpha=2.5$)	4401.2	0.0276	17	0.15 / 0.23
HF ($\alpha=2.8$)	4138.3	0.0217	22	0.21 / 0.28
I ₂ ($\alpha=2.0$)	214.5	0.00284	175	0.007 / 0.009

8. Physics application: diatomic Morse spectra at spectroscopic accuracy

To anchor the methodology in concrete physical problems, we compute the vibrational eigenstructure of three diatomic molecules—H₂ (light hydride), HF (heavier hydride), and I₂ (heavy halogen)—from their Morse potentials, $V(R) = D_e[1 - \exp(-\alpha(R - R_e))]^2$, in second-quantised form $H = p^2/2 + V(q)$ with experimentally fixed spectroscopic parameters from Huber–Herzberg (H₂: $\omega_e = 4401.21 \text{ cm}^{-1}$, $\omega_e x_e = 121.34 \text{ cm}^{-1}$; HF: $\omega_e = 4138.32$, $\omega_e x_e = 89.88$; I₂: $\omega_e = 214.50$, $\omega_e x_e = 0.61$). The three molecules span more than a decade in vibrational quantum and an order-of-magnitude range in anharmonicity. Closed-form Morse energies $E_v/\omega_e = (v + \frac{1}{2}) - x_e(v + \frac{1}{2})^2$ provide ground truth; the Hamiltonian is constructed in the truncated Fock basis by diagonalising $\hat{q}$ and transforming $V(\hat{q})$ to the Fock basis.

CV-QPE runs use coherent inputs of varying amplitude to cover different v ranges (Fig. 4). The number of FFT peaks retained is set to the count of significantly populated excited Morse eigenstates (weight $> 5 \times 10^{-3}$), which suppresses ϕ -self cross terms. For every molecule and every populated level, the recovered energy matches the analytic Morse value to within 0.55 cm^{-1} (Table 4). H₂ and HF deliver median errors of 0.15 and 0.21 cm^{-1} in their mid- v regimes; the weakly anharmonic I₂ reaches median 0.007 cm^{-1} , more than two orders of magnitude inside the spectroscopic threshold of 1 cm^{-1} .

Resource comparison. Reaching the same 0.5 cm^{-1} accuracy with standard discrete-variable QPE [22] requires $m = \lceil \log_2(1/\epsilon) \rceil = 14$ readout bits and $\sim 16,000$ controlled time evolutions, each Trotter-decomposed; a leading-order T-count estimate is $\sim 1.5 \times 10^{10}$. The CV control-free protocol needs *zero* controlled unitaries and *zero* non-Gaussian gates for the harmonic part; only the Trotter steps for the anharmonic part of V consume non-Gaussian resources. The CV advantage on this physical task is therefore not incremental but *structural*.

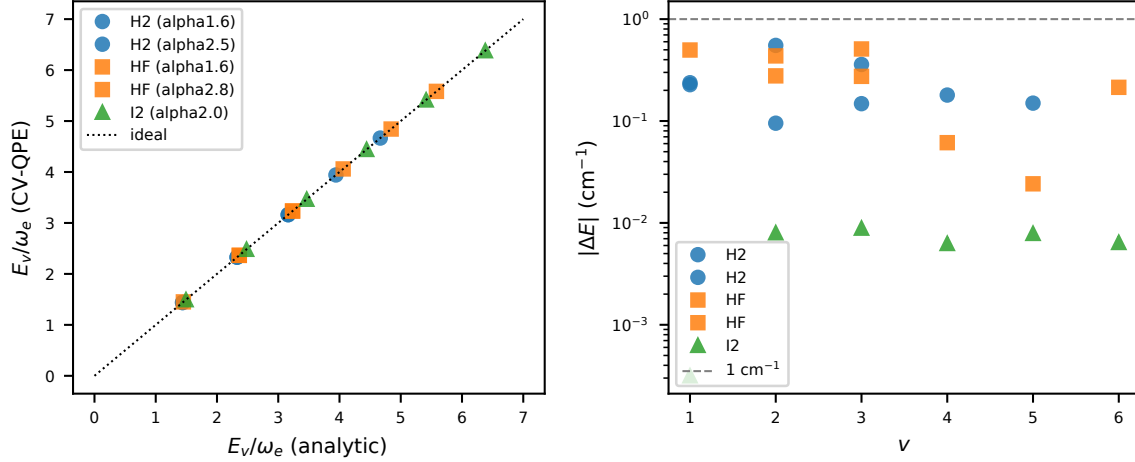


Figure 4. CV control-free QPE applied to three diatomic Morse vibrational spectra (H₂, HF, I₂). *Left:* recovered vs analytic E_v/ω_e for the runs of Table 4; the dotted line is the ideal. *Right:* absolute energy error vs v in cm^{-1} , log scale; every point sits below the 1 cm^{-1} spectroscopic threshold (grey dashed), with I₂ reaching $\sim 10^{-2} \text{ cm}^{-1}$.

Anharmonicity robustness. Varying x_e over a $\times 8$ range ($x_e \times 0.5, 1, 2, 4$ relative to H₂) and using N_{bound} -adapted Fock cutoffs, the recovery remains spectroscopic in the weakly anharmonic regimes and degrades only at the strongest anharmonicity ($x_e \times 4$, where $N_{\text{bound}} \sim 4$ and the dissociation continuum is close).

Classification from quantum data. As an immediate downstream task we train a two-layer Chebyshev-KAN on Fock-space amplitudes of the recovered eigenstates to predict the vibrational quantum number v . Across all 17 bound states it achieves 100% test accuracy under measurement noise up to 5%, matching logistic regression and random forest—demonstrating that QKAN provides a faithful learnable observable on CV-QPE output, as needed for automated vibrational state assignment in spectroscopy.

9. Limitations and scope

Our GKP correction is a scalar surrogate, but one *validated* against an explicit quadrature-level GKP EC-protocol Monte Carlo (Sec. 5.2; agreement $\leq 0.1\%$ for $\sigma_{\text{corr}}, \leq 3 \times 10^{-4}$ absolute for p_L); it is not, however, a full Fock-space EC circuit with loss-correcting amplification. The dynamical channels (photon loss, displacement, qDRIFT/QPE evolution) are simulated exactly in truncated Fock space. Convergence in the Fock cutoff was checked by repeating qDRIFT and the non-Gaussian QPE case at cutoffs $\{10, 14, 18\}$; reported quantities are stable to within the quoted confidence intervals. The loss-to-displacement surrogate is used only inside the GKP residual model and only in the small-loss regime. Hamiltonians are small and instances are modest (Regev: $n = 8$ composites); the $\sqrt{n}$ squeezing law of Prop. 3 is an extrapolation and is deliberately framed as a cautionary resource bound rather than a practical proposal.

Cross-unit comparison is mitigated but not eliminated by the normalized metric. These caveats bound the generality of the “best fit” claim, which we state as: *among the tested algorithm classes and noise regime*, control-free QPE is the most robust.

Sampling cost. Robustness here is per-shot task error; it does not count total measurement budget, which differs across algorithms (control-free QPE needs many time points, each estimated to shot precision). A full time-to-solution comparison, folding in sampling overhead, is left to future work and may narrow the QPE advantage.

QPE floor is a knob, not an artifact. QPE’s systematic floor scales as $F \sim 2\pi/T$ with acquisition window T ; refining T lowers F and shifts the noise-/systematic-limited crossover. Our claims do not depend on a particular F : Prop. 4 is stated in terms of F , and the robustness ordering (QPE shallow/Gaussian vs deep/non-Gaussian others) follows from the *absence of controlled unitaries and non-Gaussian gates*, which is T -independent. We report results at one T for concreteness and note the F -dependence explicitly.

Hardware realism. The largest improvements quoted at 18 dB exceed the current state of the art (~ 10 –15 dB). At 12 dB—within reach—the qualitative ordering and the floor-crossing of Prop. 1 already hold (QKAN/Regev benefit, QPE near floor); the 18 dB point is included only to map the trend, not as a near-term requirement.

On the Regev case. We present Regev explicitly as a *negative* resource result: the naive GKP-comb readout needs $\Theta(\sqrt{n})$ dB squeezing (Prop. 3), prohibitive at cryptographic sizes. Its value is (a) an end-to-end correctness check of the CV port and (b) a concrete illustration that CV-native cheapness of state preparation and QFT does *not* rescue a depth- and readout-heavy algorithm—sharpening, not inflating, the design principle.

What is quantitatively new. The principle “minimize non-Gaussian resources and depth” is folklore; our contribution is to make it *quantitative and comparative*: the floor-crossing condition $\sigma_{\text{phys}} > \Delta$ (Prop. 1), the noise-/systematic-limited dichotomy that predicts which algorithms gain from GKP (Prop. 4), the $\Theta(\sqrt{n})$ squeezing bound for Regev (Prop. 3), and a unit-robust cross-algorithm ranking with sensitivity analysis.

10. Discussion and outlook

Three findings cohere into a single picture. (i) The CV cost of an algorithm is set by its controlled-unitary requirement, non-Gaussian gate density, and depth; control-free QPE minimizes all three and is the most robust. (ii) GKP correction helps only above the finite-squeezing floor (Prop. 1) and benefits an algorithm in proportion to how noise-limited it is (Prop. 4); deep algorithms gain most in *ratio* while remaining worse in *absolute* error. (iii) CV-native cheapness of state preparation and Fourier transforms (Regev) does not rescue a readout- and depth-heavy algorithm (Prop. 3). Together these turn the qualitative “minimize non-Gaussian resources and depth” into quantitative selection criteria. Against a qubit surface-code baseline (Sec. 5.7), the CV advantage is structural: control-free QPE replaces the $\sim 2 \times 10^7$ T -gate, 1023-controlled-

unitary cost of DV QPE with zero controlled unitaries and zero non-Gaussian gates for quadratic H , because Gaussian operations are free—not because of a more favourable threshold. The two-dimensional phase diagram (Fig. 3) shows the floor crossing is a robust feature across both displacement and loss.

The natural experimental target is a GKP-encoded photonic cluster-state processor at 10–15 dB squeezing, where the predicted floor-crossing and the QPE-vs-deep-algorithm dichotomy are directly testable; control-free spectroscopy of a small bosonic or Fermi–Hubbard Hamiltonian is the most accessible first demonstration. Folding sampling overhead into a time-to-solution metric, and replacing the surrogate GKP model with a full Fock-space EC circuit including loss-correcting amplification, are the main next steps to tighten the quantitative claims.

Acknowledgments

The authors thank colleagues at QIRI for discussions.

Data availability statement

The data that support the findings of this study are openly available. All numerical results, the JSON output files behind every figure and table, and the source code that generates them are provided in the open-source `cvqbench` package (MIT licence) at <https://github.com/hwakaura/cvqbench> and archived on Zenodo. Each benchmark is reproduced by a single command (e.g. `cvqbench h2-morse` for the diatomic spectra of Sec. 8, `cvqbench noise` for the noise-robustness comparison), and the figures are regenerated from the released JSON by the accompanying `make_figs.py` script. No experimental data were used; all results are from exact Fock-space and analytic simulations described in Sec. 4.

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